

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	RDS-0000000001
Award Program	
Project Title	Developing a model protocol for tracking antibiotic use and AMR surveillance across a large-scale commercial swine production system: Year 2
Grant Start Date	07/01/2021
Grant End Date	06/30/2022
Total Grant Budget	\$358,660
Total FFAR Budget	\$179,330
Matching Funder(s)	National Pork Board, Pipestone Research
Final Report Date	10/13/2022
Project Director/Principal Investigator Information	
Full Name	Scott Allen Dee
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Grantee Organization Information	
Organization Name	Pipestone Research
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Authorized Signing Official (ASO) Information	
Name	Francisco Cabezon
ASO Title	Research President
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Phone	5052152852

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.
Pipestone, MN USA, 7th congressional district
- 1.2. How many new jobs were created because of this grant funding?
0
- 1.3. How many jobs were maintained because of this grant funding?
0
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
No
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Monitoring antimicrobial use (AMU) and antimicrobial resistance (AMR) on farms is recognized as an important component of antimicrobial stewardship yet the process can be resource intensive. This project has completed 2 years of a collaboration across government, academia, and a private sector veterinary practice focused on swine production in the Midwestern U.S. This

work was supported by farmers and the swine industry. Twice-annual collection of samples from pigs and the environment, along with AMU monitoring, occurred on 138 swine farms. As the longitudinal monitoring project continues, it may serve as a model for future efforts to monitor AMU and AMR. This project represents one of the first attempts to monitor AMU as well as AMR in both swine and public health pathogens on a large-scale commercial swine system in the United States. As the project is scheduled for a 5+ year period, final results are not yet available. Preliminary observations will be reported.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community.

The threat of Antimicrobial Resistance (AMR) is a major concern to human medicine, veterinary medicine, public health and livestock agriculture, as well as global society. Even though agriculture is often considered the source of AMR in humans, no data are available to support this hypothesis. To complicate the matter, while AMR data are being tracked by the National Antimicrobial Resistance Monitoring System (NARMS) at the level of the meat case (FDA), the harvest facility (USDA) and across cases of human foodborne illness (CDC), no such effort currently exists at the level of the swine farm. Therefore, the US swine industry must be proactive and begin to collect these data across pathogens of food safety significance and veterinary significance from swine farms using standardized approaches based on NARMS standards and correlate this information with antibiotic usage data for participants. This project will be the first attempt to accomplish this goal at the level of a large-scale commercial swine production system. In addition, these data will be compared in real time with antibiotic use data from participating farms using a novel approach called PART (Pipestone Antibiotic Resistance Tracker). Data from this project will serve as a model to advance the US swine industry to the global forefront in the areas of Responsible Antibiotic Use and Antimicrobial Resistance Surveillance. As this is a long-term project, final results are not yet available; however, we look forward to further years of discovery and continuing to assist the industry by improving the understanding of AMU and AMR.

2.3. Research Outcomes/Impact

The threat of Antimicrobial Resistance (AMR) is a major concern to human medicine, veterinary medicine, public health and livestock agriculture, and global society. Even though agriculture is often considered the source of AMR in humans, no data support this hypothesis. To complicate the matter, while AMR data are being tracked by the National Antimicrobial Resistance Monitoring System (NARMS) at the level of the meat case (FDA), the harvest facility (USDA) and across cases of human foodborne illness (CDC), no such effort exists at the level of the swine farm. Therefore, the US swine industry must be proactive and begin to collect these data across pathogens of food safety significance and veterinary significance from swine farms using standardized approaches based on NARMS standards and correlate this information with antibiotic usage data for participants. This project will be the first attempt to accomplish this goal at the level of a large-scale

commercial swine production system. It provides a platform for on-farm analysis, as well as lays the foundation for similar analyses to be applied at the processing plant.

2.4 What were the goals/specific aims of the project?

1. To conduct antimicrobial resistance (AMR) surveillance of pathogens of veterinary importance and food safety importance across a large-scale commercial swine production system.
2. To compare these AMR data with antibiotic use data across study farms.
3. To validate a model protocol of AMR surveillance for the US swine industry.

Timetable and completed tasks

July 2021: initiate year 2 sampling per protocol

August 2021: monitor sampling progress, submit year 2 proposal

September 2021: data presentation to stakeholders, expert panel

October 2021: monitor sampling progress

December 2021: monitor sampling progress

January 2022: Expert panel meeting, writing of mid-term report

February 2022: monitor sampling progress

April 2022: monitor sampling progress

June 2022: complete year 2, writing of final report, prepare submission for year 3 funding

2.5 Have any of the major goals/specific aims or milestones for the project changed ~~se~~ the award?

No changes have been made.

2.6 What was accomplished under the goals/specific aims or milestones for the project?

The project has now been running for two years with several more years hopefully to come. Successful outcomes so far include public: private partnership team building, data collected from farm sites enrolled in the study and isolation of targeted pathogens, preliminary statistical analysis of antimicrobial use (AMU) and AMR, use of both phenotypic and genotypic methods to evaluate AMR, and the continuous improvement of the protocol based on year 1-2 learnings. In addition, we have partnered with USDA NAHMS who has brought not only \$150,000 and \$240,000 of funding in years 2 and 3 respectively, to project, but significant expertise in data analysis. Preliminary observations (year 1 vs year 2) include completion of sampling across enrolled sites, similar recovery rates of targeted bacteria at the level of the pig and the environment, and no significant differences in AMR between sick and healthy pigs across all organisms tested. Regarding efforts in whole genome sequencing, the serovar prediction of *E. coli* and *Salmonella* genomes revealed high diversity indicating that the no strains were endemic. However, since this analysis contained only 69 *E. coli* isolates and 170 *Salmonella* isolates, further years of sampling is critical to grow the database. Finally, we are beginning to identify outliers (problem farms), i.e., farms with high AMU, or farms with a higher prevalence of monophasic *Salmonella typhimurium*. Clearly, we need to understand why these observations are occurring. Is it due to farm health status, animal flow, building design and environment, recent disease

challenges, etc? As we develop our database, we will be able to tease apart these outliers, determine causation, apply interventions, and measure outcomes. Therefore, further sampling is critical. This is the innate value of this project, to identify problems and solve them for farmers and consumers. Due to its unique design and resource base, there is not another project that can do this. We look forward to further years of discovery and continuing to assist the industry by improving the understanding of AMU and AMR.

- 2.7 Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Pipestone has been organizing this plan since 2016, including the development of the PART program and the validation of environmental sampling protocols. We collaborate with scientists from SDSU, as well as the USDA, to conduct all independent laboratory evaluations. In addition, the PI is board certified in bacteriology by the American College of Veterinary Microbiologists. Finally, successful completion of year 1 and 2 has given us confidence in our data integrity and repeatability of our protocol.

- 2.8 List key stakeholders that could be served by results of this project.
Stakeholder groups to be served include pork producers, livestock agriculture, veterinarians, scientists, human medical and public health personnel, and government officials.

- 2.9 How have the results of this reporting period been disseminated to communities of interest?

This information was communicated via peer-reviewed publications, white papers, briefs, the Pipestone PART website, and educational bulletins across the nation at continuing educational meetings targeting pork producers, livestock agriculture, veterinarians, scientists, human medical and public health personnel, and government officials. Specific sessions where data were presented included American Veterinary Medical Association annual meeting, National Institute for Animal Agriculture, Iowa State University McKean lecture and the Minnesota Veterinary Medical Association. In addition, a publication describing year 1 data was submitted to the OIE journal and review is in process.

- 2.10 Describe challenges or delays encountered during project and actions that were taken to resolve them.

No delays were experienced.

- 2.11 What opportunities for training and professional development has the project provided during this reporting period?

Nothing to Report

- 2.12 Please indicate the number of undergraduate and graduate students, post- doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail.

Nothing to report

3 Information Products

- 3.6 Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Conference proceedings: AVMA, MVMA, CDC, NIAA, AASV, KSU, ISU

- 3.7 Please provide a list of citations for the information products produced during the project.

Due to the large number of citations, a comprehensive list can be provided if requested.

- 3.8 Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award?

Antimicrobial use & resistance monitoring for pig production in the United States. T. Spronk, A. L. Green, M. Vuolo, L. Ruesch, R. Edler, C. Haley, J. Scaria, J. Hennings, S. Dee, & C.B. Shivley, *Rev. Sci. Tech. Off. Int Epiz* (under review)

- 3.9 Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Not applicable at this time.

- 3.10 Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

Not applicable

- 3.11 Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Not applicable

- 3.12 Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

All isolates and WGS associated with this project are stored at South Dakota State University and are fully accessible.

4 Data Management

- 4.6 Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

We have saved every bacterial isolate cultured from this project at -80 C, all whole genome sequences and all antimicrobial susceptibility testing results. These data are all stored at South Dakota State University Animal Disease Research and Diagnostic Laboratory in Brookings, SD USA.

- 4.7 If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

All are related to each other as they were generated via on-farm sampling efforts associated with the project.

- 4.8 Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Not applicable



Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Scott Dee

PI Signature:  Date:

10/20/22

Authorized Signing Official (ASO) Name: FRANCISCO CABEZA

ASO Signature:  Date: 10/20/22